MoonWalker Weekly Meeting

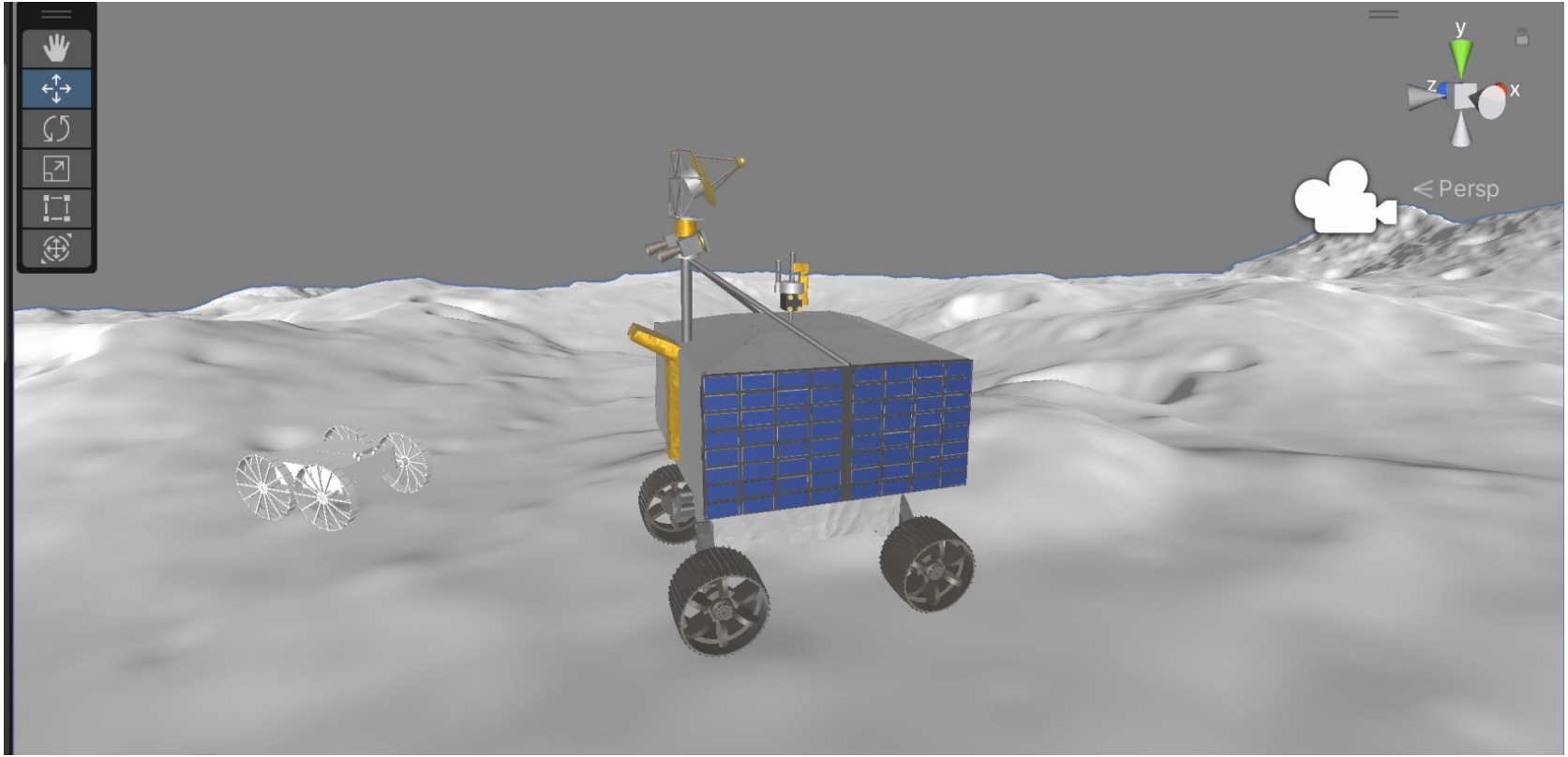
10/16-10/22

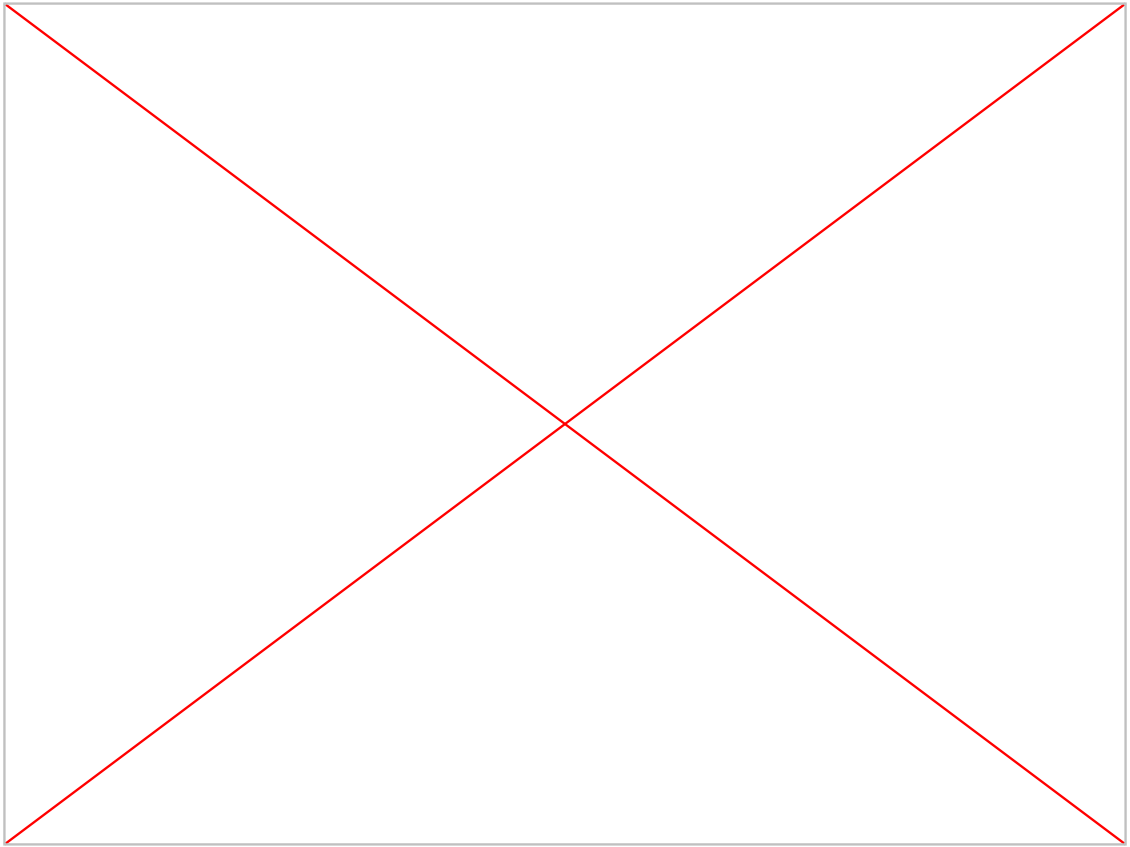
Rowel, Martin, Elijah

Progress Update

- Martin
  - Set up Unity connection with HSML API over network
  - Set up movement controls for Viper rover
- Elijah
  - Set up Omniverse connection with HSML API:
    - Validated Tracker script: produces JSON file with coordinates
    - Connected to HSML API as consumer and producer (able to send and receive messages)
- Rowel
  - Switch to the rework branch of HSML-Form-Website so that we can continue testing
  - Collaborate with group mates to begin testing

Unity





Isaac Sim Full 5.0.0 - file:/D:/School/Research/Current Research/NASA-JPL Digital Twin Interopability/Omniverse Isaac Sim/Models/MAINDEMO.usd*

File Edit Create Window Tools Utilities Layouts Help

LIVE CACHE: NEW VERSION DETECTED

RTX - Real-Time | Perspective | Default

FPS: 115.50, Frame time: 8.66 ms
NVIDIA GeForce RTX 4070 Laptop GPU: 3.7 GiB used, 3.4 GiB available
Process Memory: 1.7 GiB used, 5.8 GiB available
1280x720

Stage | Layer | Render Settings

Search

Name (Old to New)	Type
World (defaultPrim)	Xform
PhysicsScene	PhysicsScene
Looks	Scope
Shackleton_Ridge_300x300_v4_	Xform
Unity_Viper	Xform

C:\WINDOWS\system32\cmd.

```
Error reading or updating prim: 'FileReaderScript' object has no attribute 'file_path'
[INFO] Wrote new transform data to C:\Users\elija\AppData\Local\Temp\producerOmniUpdate.json
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```

Console | Content

Import | D:/School/Research/Current Research/NASA-JPL Digital Twin Interopability/Omniverse Isaac Sim/Models/

- Setup Documenta
- Papers
- RAGGY
- Timesheetes
- Unity_HSML
- My Published Papers
- Past Research
- Research Paper To Read
- Resume
- SteamLibrary
- Workouts

4WD_Contrl.usd | BehaviorScript_pr...ce.py | CADREDemo.usd | MAINDEMO.usd | Shackleton Ridg....usd

File size 4.00 KB

File Information

No Additional Information

Checkpoints

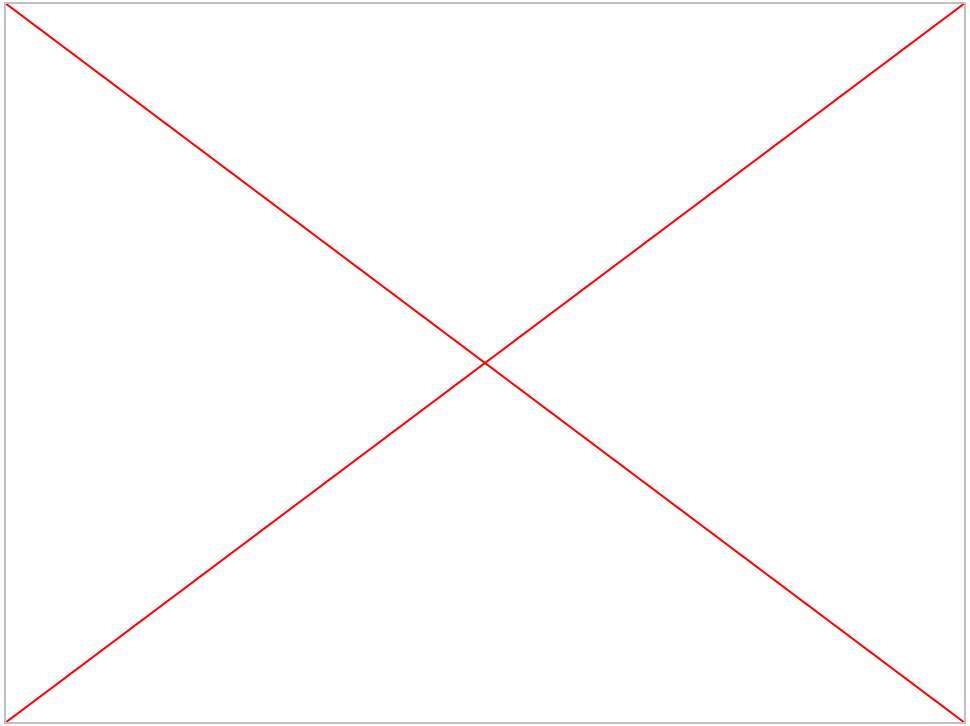
Geometry

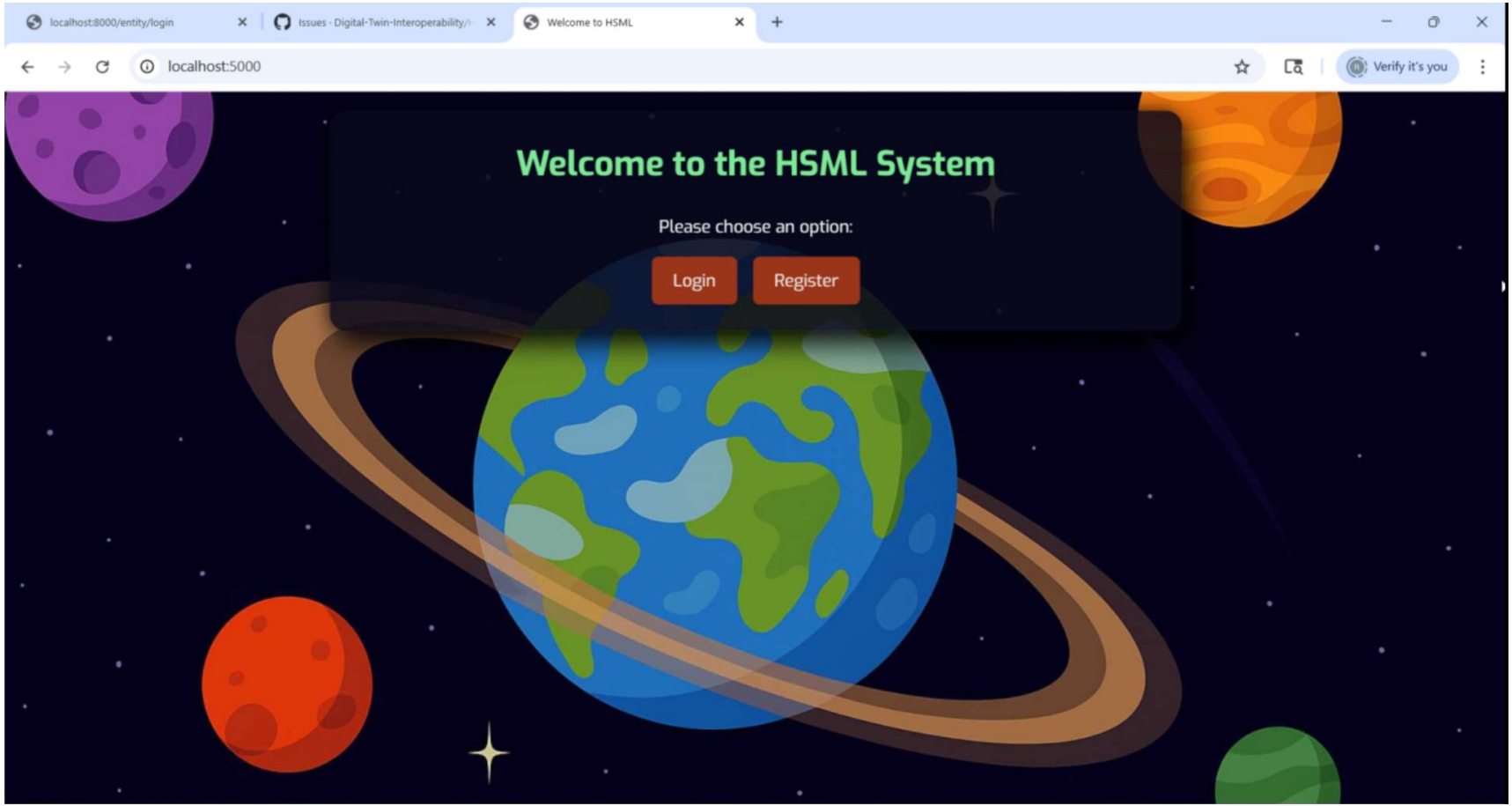
Mesh

Normals [...]

Orientation rightHanded

Points [...]




localhost:8000/entity/login
Issues · Digital-Twin-Interoperability/
Welcome to HSML
localhost:5000
Verify it's you
Welcome to the HSML System
Please choose an option:
Login
Register

Challenges

- Registration of Users in the rework branch of the HSML-Form-Website
  - Users are redirected to the registration page again after registering (no downloadable pem key or user registered in MySQL)

Project | Console

Clear | Collapse | Error Pause | Editor

1 6 1

[08:48:01] Authentication failed: HTTP/1.1 403 Forbidden
UnityEngine.Debug:LogError (object)

```
Authentication failed: HTTP/1.1 403 Forbidden
UnityEngine.Debug:LogError (object)
unityApiProducer/<AuthenticateWithAPI>d__13:MoveNext () (at Assets/Scripts/unityApiProducer.cs:131)
UnityEngine.SetupCoroutine:InvokeMoveNext (System.Collections.IEnumerator,intptr)
```

```
Windows PowerShell

(venv) PS D:\School\Research\Current Research\NASA-JPL Digital Twin Interopability\
hsml_schema\scripts\BL_plugins_HSML_API\Omni> python ViperAOmniAPIJSON.py
{'detail': 'Authorization failed.'}
{'detail': 'User not authorized for this topic.'}
Polling latest messages from HSML API. Press Ctrl+C to stop.

No new message yet or error fetching.
No new message yet or error fetching.
```

```
(venv) PS D:\School\Research\Current Research\NASA-JPL Digital Twin Interopability\hsml_
schema\scripts\BL_plugins_HSML_API\Omni> python CadreAOmniAPIJSON.py
{'detail': 'Authentication failed.'}
Message sent: {'detail': 'User not authenticated for this topic.'}
Message sent: {'detail': 'User not authenticated for this topic.'}
Message sent: {'detail': 'User not authenticated for this topic.'}
Message sent: {'detail': 'User not authenticated for this topic.'}
```

Action Plan

- Martin
  - Run test 1 and 2 (shared goal)
  - Work with Elijah on collision
- Elijah
  - Run test 1 and 2 (shared goal)
  - Create new BehaviorScripts and explore collision/interaction
- Rowel
  - Run test 1 and 2 (shared goal)
  - Check the connections between the reworked branch of the HSML-Form-Website with Kafka and MySQL

Omniverse sending data

```
(venv) PS D:\School\Research\Current Research\NASA-JPL Digital Twin Interopability\hsml_schema\scripts\BL_plugins_HSML_API\Omni>
(venv) PS D:\School\Research\Current Research\NASA-JPL Digital Twin Interopability\hsml_schema\scripts\BL_plugins_HSML_API\Omni> python CadreAOmniAPIJSON.py
Authentication:
{'message': 'Authentication successful for topic omni_cadrea_producer_t9glwc'}
Connecting to get topic
Message sent: {'message': 'Message sent successfully'}
Message sent: {'message': 'Message sent successfully'}
Message sent: {'message': 'Message sent successfully'}
Message sent: {'message': 'Message sent successfully'}
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